

SUMMARY (narrative by template; all figures from the deterministic engine)

The request for Rs 180,000 over 48 months is referred for manual review under the default policy (v1.0). Verified monthly income is Rs 38,300 against existing obligations of Rs 3,000; the proposed EMI of Rs 5,010 gives a FOIR of 20.9% (limit 50.0%) and an LTV of 75.0% (limit 80.0%). Risk band: LOW. Reason codes: INCOME_MISMATCH.

For the borrower:

Priya Sharma, your application for Rs 180,000 needs a manual check before a decision. Some details in your documents did not fully agree, so an underwriter will review them with you.

Key drivers:

- FOIR 20.9% vs limit 50.0%
- LTV 75.0% vs limit 80.0%
- Verified income Rs 38,300 (volatility 3.5%)
- INCOME_MISMATCH: Monthly income differs across documents by 15.6%

Next steps:

- Underwriter to resolve the reconciliation flags with the borrower.
- Re-run the workflow once corrected documents are uploaded.

1. DECISION

Decision: REFERRED Confidence: 0.65

Reason codes: INCOME_MISMATCH

- [warning] INCOME_MISMATCH: Monthly income differs across documents by 15.6%

1b. WHY THIS DECISION (policy margins + SHAP; words by template)

Priya Sharma's loan is referred to a human underwriter because Monthly income differs across documents by 15.6%.

No hard limit was breached, so an underwriter can still approve it after resolving the flags.

The model's risk band is LOW (score 0.3038 on a 0-1 scale where 0 is safest; a typical synthetic borrower scores 0.437336).

What pushes risk up: ltv (loan / asset value) (75.0% vs a typical 68.1%, +0.012); repayment history score (70.0% vs a typical 71.7%, +0.003); income mismatch across documents (15.6% vs a typical 12.7%, +0.003).

What pulls risk down: foir (obligations / income) (20.9% vs a typical 41.1%, -0.061); emi-to-income ratio (13.1% vs a typical 34.4%, -0.032); employment stability (96.5% vs a typical 71.8%, -0.025).

The risk score is a secondary signal; the policy rules above decide the outcome.

Mathematics:

PASS FOIR 20.9% vs limit 50.0% (-29.1 pp under)

PASS LTV 75.0% vs limit 80.0% (-5.0 pp under)

PASS age 31 vs minimum 21 (+10 years)

PASS verified monthly income Rs 38,300

risk score 0.303766 = base 0.437336 + sum SHAP -0.13357 (method additive_exact, additivity error 0.0)

phi(foir) = -0.0605 [value 20.9% vs baseline 41.1%]

phi(emi_to_income) = -0.0320 [value 13.1% vs baseline 34.4%]

phi(employment_stability) = -0.0247 [value 96.5% vs baseline 71.8%]

phi(income_volatility) = -0.0189 [value 3.5% vs baseline 22.4%]

phi(ltv) = +0.0124 [value 75.0% vs baseline 68.1%]

2. BORROWER SUMMARY (declared facts)

ref_id: REF-APP-MISMATCH-002

full_name: Priya Sharma

age: 31

location_tier: tier_2

employment_type: gig_platform

declared_monthly_income: 45,000.00

loan_request.amount: 180,000.00

loan_request.down_payment: 60,000.00

loan_request.tenure_months: 48

loan_request.vehicle_price: 240,000.00

3. EXTRACTED FACTS (evidence with citations)

full_name = Priya Sharma conf 0.95 [application_form p.None] auto
monthly_income = 45,000.00 conf 0.95 [application_form p.None] auto
loan_amount = 180,000.00 conf 0.95 [application_form p.None] auto
vehicle_price = 240,000.00 conf 0.95 [application_form p.None] auto
full_name = Priya Sharma conf 0.85 [kyc.txt p.1] auto
identity = XXXX-XXXX-8765 conf 0.9 [kyc.txt p.1] auto
address = 22 Lake View, Bengaluru 560001 conf 0.85 [kyc.txt p.1] auto
monthly_income = 38,500.00 conf 0.92 [bank_statement.txt p.1] auto
bank_credits = [38500.0, 37200.0, 39900.0, 38100.0, 36800.0, 40500.0] conf 0.9 [bank_statement.txt p.1] auto
existing_monthly_obligations = 3,000.00 conf 0.92 [bank_statement.txt p.1] auto
address = 22 Lake View, Bengaluru 560001 conf 0.85 [bank_statement.txt p.1] auto
vehicle_price = 240,000.00 conf 0.92 [dealer_invoice.txt p.1] auto
loan_amount = 180,000.00 conf 0.92 [dealer_invoice.txt p.1] auto
platform_income = 38,000.00 conf 0.92 [platform_earnings.txt p.1] auto
monthly_income = 38,000.00 conf 0.92 [platform_earnings.txt p.1] auto

4. CALCULATED VALUES (deterministic engine, formula v1)

verified_income: 38,300.00
existing_obligations: 3,000.00
emi: 5,009.53
foir: 0.2091
ltv: 0.75
income_volatility: 0.0347
income_stability: 0.9653
loan_to_income: 0.3916
emi_to_income: 0.1308
input.annual_rate: 15.00
input.existing_monthly_obligations: 3,000.00
input.loan_amount: 180,000.00
input.monthly_incomes: [38500.0, 37200.0, 39900.0, 38100.0, 36800.0, 40500.0]
input.tenure_months: 48
input.vehicle_price: 240,000.00

5. RISK SIGNAL (secondary, never decides)

score 0.3038 band LOW model xgb-v1/fallback

6. AI REASONING

strengths: FOIR 21% is comfortably serviceable
strengths: LTV 75% leaves borrower equity cushion
risks: INCOME_MISMATCH: Monthly income differs across documents by 15.6%
reasoning: Deterministic FOIR 20.91%, LTV 75.00%; XGBoost band LOW (score 0.3038).

7. POLICY

profile default version 1.0
limit.foir_max: 0.5
limit.ltv_max: 0.8
limit.max_tenure_months: 60
limit.min_age: 21

8. HUMAN OVERRIDES

none

SYNTHETIC DATA - demo only. Financial numbers computed deterministically; LLM never sets numbers.